

Modular Magnetic Headphones

Engineering Notebook

Project ID: ENG-001 Version: 0.1

Project Dashboard

Project Name: Modular Magnetic Headphones
Project ID: ENG-001
Version: 0.1
Status: **Concept**
Category: Consumer Electronics / Hardware Design
Started: 17 May 2025
Last Updated: 2026-06-22

1 Project Overview

Idea

Design a semi-modular headphone system with retractable magnetic headbands, power/data-sharing connectors, ANC synchronisation, and an interactive earcup screen.

2 Engineering Questions

Engineering Question

Can magnetic connectors safely support mechanical attachment, charging, and data transfer without compromising comfort or reliability?

3 Design Decisions

Design Decision

Date: 2026-06-22
Decision: Explore MagSafe-style connector points between the retractable band and earcups.
Reason: This may simplify attachment while supporting modularity.
Trade-off: Magnetic alignment may introduce cost and durability challenges.

4 Experiments

Experiment

Date: 2026-06-22

Objective: Test whether a retractable rail can fit inside the earcup housing.

Method: Begin with hand sketches, then produce a simplified 3D model.

Success Rating: 4/10

Confidence Rating: 5/10

Worth Pursuing: Yes

5 Failures and Problems

Failure / Problem

Date: 2026-06-22

Problem: The first headband concept may require more internal space than expected.

Observation: The earcup may become too bulky if the rail mechanism is not compact.

Next Step: Compare telescopic, flexible-band, and folding-arm mechanisms.

6 Discoveries

Discovery

Date: 2026-06-22

Context: Thinking about modular headphones.

Observation: A magnetic connector might serve three roles: alignment, attachment, and electrical contact.

Potential Applications:

- Modular headbands
- Swappable earcups
- Charging docks

7 Sketches

Sketch Placeholder

Draw the retractable headband concept here.

8 Engineering Calculation

Engineering Calculation

Purpose: Estimate the holding force needed for the magnetic connector.

Inputs: Estimated headphone mass, expected movement, contact area.

Assumptions: The user will experience normal walking and head movement only.

Calculation: To be added.

Interpretation: To be added after first estimate.

9 Future Research Queue

Future Research Queue

Investigate:

- Hall-effect sensors
- Magnetic pogo pins
- Flexible PCB connectors
- Bluetooth antenna placement

10 Note to Future Self

Note to Future Self

Do not redesign the entire headphone too early. First prove the retractable mechanism, then test power/data connection, then think about aesthetics.

11 What I Wish I Knew Earlier

What I Wish I Knew Earlier

Mechanical feasibility should be checked before adding complex electronics. A beautiful product idea can fail if the housing cannot physically contain the mechanism.